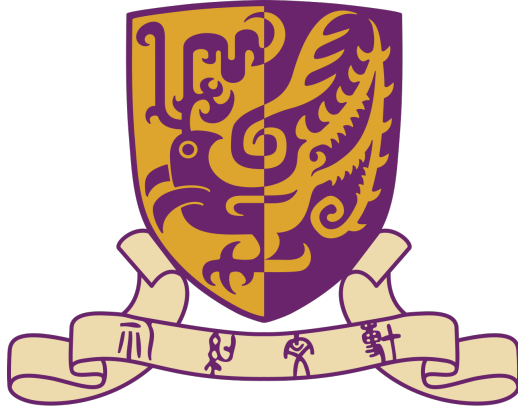




THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN

INDEPENDENT REPORT

MATRIX ANALYSIS

Singular Values and Integer Powers of Accretive Matrices

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Abstract

This report studies inequalities comparing the singular values of an accretive matrix with those of its positive integer powers. Throughout, $k \in \mathbb{N}$ and $k \geq 2$. For every corresponding pair of singular values, we prove the dimension-free estimate

$$\sigma_j(A)^k \leq 2^{k(k-1)/2} \sigma_j(A^k), \quad j = 1, \dots, n.$$

Thus a universal constant exists for every fixed integer k . However, no single constant works simultaneously for all $k \geq 2$. We also give lower and upper bounds for the best constants and explain the special cases of the largest and smallest singular values.

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1 Problem statement and conventions

Throughout, $\mathbb{N} = \{1, 2, 3, \dots\}$.

Definition 1.1 (Accretive matrix). Let $n \in \mathbb{N}$. A matrix $A \in \mathbb{R}^{n \times n}$ is *accretive* if

$$x^T A x \geq 0 \quad \text{for every } x \in \mathbb{R}^n.$$

It is *strictly accretive* if $x^T A x > 0$ for every nonzero $x \in \mathbb{R}^n$.^a

(For background on accretive matrices through the theory of numerical ranges, see [3].)

^aAt the beginning I use the definition $x^T A x > 0$. Later I found that we can give some relaxation of the condition as $x^T A x \geq 0$

Definition 1.2 (Singular values). For a matrix $B \in \mathbb{R}^{n \times n}$, write its singular values in nonincreasing order as

$$\sigma_1(B) \geq \dots \geq \sigma_n(B) \geq 0,$$

and denote $\sigma(B) = \sigma_1(B)$. Standard references for singular values, variational principles, and matrix inequalities include [1, 4].

For vectors $x \in \mathbb{R}^n$, the notation $\|x\|_2$ denotes the Euclidean norm. For matrices B , the largest singular value is consistently written as $\sigma(B)$, as specified above. In this report every exponent satisfies

$$k \in \mathbb{N}, \quad k \geq 2.$$

We investigate the following three questions, in this order:

1. Is there a universal constant C such that, for every $n \in \mathbb{N}$ and every accretive matrix $A \in \mathbb{R}^{n \times n}$, the inequality

$$\sigma(A)^2 \leq C \sigma(A^2)$$

holds?

2. Is there a universal constant C^+ such that, simultaneously for all integers $k \geq 2$, all $n \in \mathbb{N}$, and all accretive matrices $A \in \mathbb{R}^{n \times n}$, the inequality

$$\sigma(A)^k \leq C^+ \sigma(A^k)$$

holds?

3. If the answer to Question 2 is negative, then for every fixed integer $k \geq 2$, does there exist a constant $C_k^+ = C^+(k)$, independent of n and A , such that

$$\sigma(A)^k \leq C_k^+ \sigma(A^k)$$

holds?

2 Main results

Theorem 2.1 (Answer to Question 1). *For every $n \in \mathbb{N}$ and every accretive matrix $A \in \mathbb{R}^{n \times n}$,*

$$\sigma(A)^2 \leq 2\sigma(A^2). \quad (1)$$

Thus the universal constant $C = 2$ is admissible.

Proof. See [Section 3](#).

Theorem 2.2 (Answer to Question 2). *There is no finite constant C^+ such that*

$$\sigma(A)^k \leq C^+ \sigma(A^k) \quad (2)$$

holds simultaneously for all integers $k \geq 2$, all $n \in \mathbb{N}$, and all accretive matrices $A \in \mathbb{R}^{n \times n}$. This failure occurs even for one fixed strictly accretive 2×2 matrix.

Proof. See [Section 4](#).

Theorem 2.3 (Answer to Question 3). *For every fixed integer $k \geq 2$, every $n \in \mathbb{N}$, and every accretive matrix $A \in \mathbb{R}^{n \times n}$,*

$$\sigma(A)^k \leq 2^{k(k-1)/2} \sigma(A^k). \quad (3)$$

Consequently, one may take

$$C_k^+ = C^+(k) = 2^{k(k-1)/2}.$$

Proof. See [Section 5](#).

3 Proof of Theorem 2.1

Lemma 3.1 (Kato's vector inequality). *For every accretive matrix A and every $x \in \mathbb{R}^n$,*

$$\|Ax\|_2^2 \leq 2\|x\|_2\|A^2x\|_2. \quad (4)$$

Proof. Fix $t > 0$. Direct expansion gives

$$\begin{aligned} & \|x\|_2^2 + t^4\|A^2x\|_2^2 - t^2\|Ax\|_2^2 - \|x - tAx + t^2A^2x\|_2^2 \\ &= 2t(x - tAx)^T A(x - tAx) \geq 0. \end{aligned}$$

Consequently,

$$\|Ax\|_2^2 \leq t^{-2}\|x\|_2^2 + t^2\|A^2x\|_2^2.$$

If $A^2x = 0$, letting $t \rightarrow \infty$ gives $Ax = 0$. Otherwise choose $t^2 = \|x\|_2/\|A^2x\|_2$ to obtain (4). $\square$

The estimate above is the finite-dimensional accretive counterpart of Kato's inequality [6]. Indeed, applying the dissipative formulation in [2, Theorem 2.5] to $-A$ yields the same estimate. For general background on linear operators in Hilbert spaces, see [5].

For the largest singular value, let x_* be a unit right singular vector of A corresponding to $\sigma(A)$. Applying (4) to this particular vector, and then using the definition of $\sigma(A^2)$, gives

$$\sigma(A)^2 = \|Ax_*\|_2^2 \leq 2\|A^2x_*\|_2 \leq 2\sigma(A^2).$$

The maximizing vectors for A and A^2 need not coincide; the last inequality only uses that $\sigma(A^2)$ is the maximum of $\|A^2x\|_2$ over all unit vectors. More generally, recall the singular-value min-max formula [1, 4]:

$$\sigma_j(B) = \min_{\substack{E \subseteq \mathbb{R}^n \text{ a linear subspace} \\ \dim E = n-j+1}} \max_{\substack{x \in E \\ \|x\|_2=1}} \|Bx\|_2.$$

Fix an admissible subspace E_* at which the minimum in the min-max formula for $\sigma_j(A^2)$ is attained. Then (4) implies

$$\sigma_j(A)^2 \leq \max_{\substack{x \in E_* \\ \|x\|_2=1}} \|Ax\|_2^2 \leq 2 \max_{\substack{x \in E_* \\ \|x\|_2=1}} \|A^2x\|_2 = 2\sigma_j(A^2).$$

This proves the stronger estimate

$$\sigma_j(A)^2 \leq 2\sigma_j(A^2), \quad j = 1, \dots, n. \quad (5)$$

The case $j = 1$ is (1), so Theorem 2.1 follows.

4 Proof of Theorem 2.2

Consider the fixed matrix

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = I + N, \quad N^2 = 0.$$

Its symmetric part has eigenvalues $1/2$ and $3/2$, so J is strictly accretive. For every integer $k \geq 2$, the binomial theorem gives

$$J^k = (I + N)^k = I + kN = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

Direct calculation gives

$$\sigma(J) = \varphi := \frac{1 + \sqrt{5}}{2}, \quad \sigma(J^k) = \frac{\sqrt{k^2 + 4} + k}{2}.$$

Therefore

$$\frac{\sigma(J)^k}{\sigma(J^k)} = \frac{2\varphi^k}{\sqrt{k^2 + 4} + k} \sim \frac{\varphi^k}{k} \rightarrow \infty \quad (k \rightarrow \infty, k \in \mathbb{N}).$$

Thus no constant can be independent of the integer exponent k . This proves [Theorem 2.2](#).

5 Proof of Theorem 2.3

Although [Theorem 2.2](#) has a negative conclusion, fixing the exponent changes the situation. The square estimate from [Section 3](#) can be iterated to give a constant depending only on k .

Proposition 5.1. *For every integer $k \geq 2$ and every vector $x \in \mathbb{R}^n$,*

$$\|Ax\|_2^k \leq 2^{k(k-1)/2} \|x\|_2^{k-1} \|A^k x\|_2. \quad (6)$$

Consequently,

$$\sigma_j(A)^k \leq 2^{k(k-1)/2} \sigma_j(A^k), \quad j = 1, \dots, n. \quad (7)$$

Proof. We argue by induction on k . The case $k = 2$ is [Lemma 3.1](#). Suppose $k \geq 3$ and the assertion holds with $k - 1$ in place of k . Applying the induction hypothesis to the vector Ax gives

$$\|A^2 x\|_2^{k-1} \leq 2^{(k-1)(k-2)/2} \|Ax\|_2^{k-2} \|A^k x\|_2.$$

Raise (4) to the power $k - 1$ and substitute:

$$\begin{aligned} \|Ax\|_2^{2k-2} &\leq 2^{k-1} \|x\|_2^{k-1} \|A^2 x\|_2^{k-1} \\ &\leq 2^{k-1+(k-1)(k-2)/2} \|x\|_2^{k-1} \|Ax\|_2^{k-2} \|A^k x\|_2 \\ &= 2^{k(k-1)/2} \|x\|_2^{k-1} \|Ax\|_2^{k-2} \|A^k x\|_2. \end{aligned}$$

If $Ax = 0$ there is nothing to prove. Otherwise divide by $\|Ax\|_2^{k-2}$ to obtain (6).

It remains to pass from vectors to singular values. Fix a subspace $E_* \subseteq \mathbb{R}^n$ of dimension $n - j + 1$ at which the minimum in the min-max formula for $\sigma_j(A^k)$ is attained. Then (6) gives

$$\sigma_j(A)^k \leq \max_{\substack{x \in E_* \\ \|x\|_2=1}} \|Ax\|_2^k \leq 2^{k(k-1)/2} \max_{\substack{x \in E_* \\ \|x\|_2=1}} \|A^k x\|_2 = 2^{k(k-1)/2} \sigma_j(A^k).$$

Thus the corresponding estimate holds for every singular-value index. The case $j = 1$ gives (3), proving [Theorem 2.3](#) with $C_k^+ = 2^{k(k-1)/2}$. $\square$

Remark 5.2 (The powers need not be accretive). The induction always applies the same accretive operator A to different vectors; it does not assume that $A^2, A^3, \dots$ are accretive. Such an assumption would be false. For example,

$$A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}, \quad \frac{A + A^T}{2} = I, \quad \frac{A^2 + (A^2)^T}{2} = -3I.$$

5.1 Growth of the best constants

For every integer $k \geq 2$, define the best constant over the original strictly accretive class by

$$C_{\text{opt}}(k) := \sup_{\substack{n \in \mathbb{N}, A \in \mathbb{R}^{n \times n} \\ (A+A^T)/2 \succ 0}} \frac{\sigma(A)^k}{\sigma(A^k)}.$$

Proposition 5.3. *For every integer $k \geq 2$,*

$$\boxed{\frac{k!}{3^{k/2}} \leq C_{\text{opt}}(k) \leq 2^{k(k-1)/2}}. \quad (8)$$

In particular, no fixed $M > 0$ can satisfy $C_{\text{opt}}(k) \leq M^k$ for every integer $k \geq 2$.

Proof. Let S_n have ones on the first superdiagonal and zeros elsewhere, and set

$$U_n = (I - S_n)^{-1}, \quad (U_n)_{ij} = \begin{cases} 1, & i \leq j, \\ 0, & i > j. \end{cases}$$

If $\mathbf{1} = (1, \dots, 1)^T$, then

$$\frac{U_n + U_n^T}{2} = \frac{I + \mathbf{1}\mathbf{1}^T}{2} \succ 0.$$

For $e = \mathbf{1}/\sqrt{n}$,

$$\sigma(U_n) \geq \|U_n e\|_2 = \left(\frac{(n+1)(2n+1)}{6} \right)^{1/2}.$$

Since S_n is nilpotent, the negative binomial expansion is finite:

$$U_n^k = (I - S_n)^{-k} = \sum_{\ell=0}^{n-1} \binom{k+\ell-1}{\ell} S_n^\ell.$$

All coefficients are positive. The largest row sum and largest column sum both equal

$$\sum_{\ell=0}^{n-1} \binom{k+\ell-1}{\ell} = \binom{n+k-1}{k}.$$

The largest row sum and largest column sum bound the largest singular value by their geometric mean [1, 4]. Since these two sums coincide here, $\sigma(U_n^k) \leq \binom{n+k-1}{k}$. Therefore,

$$C_{\text{opt}}(k) \geq \left(\frac{(n+1)(2n+1)}{6} \right)^{k/2} \binom{n+k-1}{k}^{-1}.$$

Letting $n \rightarrow \infty$ and using $\binom{n+k-1}{k} \sim n^k/k!$ proves the lower bound. The upper bound is (3). Finally, Stirling's formula gives

$$C_{\text{opt}}(k)^{1/k} \geq \frac{(k!)^{1/k}}{\sqrt{3}} \sim \frac{k}{e\sqrt{3}} \rightarrow \infty.$$

□

5.2 Special cases and limitations

5.2.1 The smallest singular value

For every square matrix, without any accretivity assumption,

$$\|Ay\|_2 \geq \sigma_n(A)\|y\|_2.$$

Repeated application with $y = A^{k-1}x, A^{k-2}x, \dots, x$ gives

$$\|A^k x\|_2 \geq \sigma_n(A)^k \|x\|_2.$$

Minimizing over unit vectors yields

$$\boxed{\sigma_n(A)^k \leq \sigma_n(A^k)}. \quad (9)$$

Hence the stronger constant 1 works for the smallest singular value for every integer $k \geq 2$.

5.2.2 Normal and unrestricted matrices

If A is normal, meaning $A^T A = A A^T$, then for every integer $k \geq 2$,

$$\sigma_j(A^k) = \sigma_j(A)^k, \quad j = 1, \dots, n.$$

The growth of the constants in the general theory therefore reflects the nonnormality of accretive matrices; see [1, 4] for background on normal and nonnormal matrices. If accretivity is removed, no largest-singular-value constant exists even for a fixed integer $k \geq 2$. Indeed,

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

satisfies $N^k = 0$ and $\sigma_1(N) = 1$. Even restricting to invertible matrices with positive eigenvalues does not repair the statement. For $J_t = I + tN$,

$$\frac{\sigma(J_t)^k}{\sigma(J_t^k)} \sim \frac{t^{k-1}}{k} \longrightarrow \infty \quad (t \rightarrow \infty).$$

5.3 Conclusion

For every integer $k \geq 2$, accretivity gives the dimension-free estimate

$$\sigma_j(A)^k \leq 2^{k(k-1)/2} \sigma_j(A^k)$$

for every corresponding singular-value index j . The proof follows from one elementary vector inequality and induction on k . Thus a universal constant exists when the integer exponent is fixed. A fixed strictly accretive 2×2 Jordan block shows that no constant can be uniform in k , while a family of upper-triangular accretive matrices shows that the best constants grow faster than exponentially. For the smallest singular value, the much stronger constant 1 works without accretivity.

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AI Use Disclosure

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